



Advances and Challenges in Thermal–Hydraulic Enhancement of Crossflow and Compact Heat Exchangers: A Four-Decade Review

Zahraa Asaad Abdul Hammed¹, Ehsan Fadhil Abbas^{1*}, Abdulrazzak Akroot²

¹ Mechanical Power Techniques Engineering Department, Technical Engineering College–Kirkuk, Northern Technical University, 3006 Kirkuk, Iraq

² Department of Mechanical Engineering, Faculty of Engineering, Karabük University, 78050 Karabük, Türkiye

* Correspondence: Ehsan Fadhil Abbas (ehsanfadhil@ntu.edu.iq)

Received: 02-15-2026

Revised: 04-15-2026

Accepted: 05-01-2026

Citation: Z. A. A. Hammed, E. F. Abbas, and A. Akroot, “Advances and challenges in thermal–hydraulic enhancement of crossflow and compact heat exchangers: A four-decade review,” *Int. J. Energy Prod. Manag.*, vol. 11, no. 2, pp. 304–322, 2026. <https://doi.org/10.56578/ijepm110203>.



© 2026 by the author(s). Licensee Acadlore Publishing Services Limited, Hong Kong. This article can be downloaded for free, and reused and quoted with a citation of the original published version, under the CC BY 4.0 license.

Abstract: Cross-flow heat exchangers are widely used in thermal and energy systems for their compactness and structural simplicity; however, their thermal–hydraulic performance remains strongly constrained by geometric configuration, flow regime, and pressure-drop penalties. This review systematically examines more than four decades of research on cross-flow and compact heat exchangers, covering theoretical, numerical, and experimental investigations. The effects of geometric modifications—such as fin and tube shape, pitch, orientation, and surface interruption—are critically analyzed, revealing that non-uniform, flow-disturbing geometries can enhance heat transfer by 15–50%, albeit often at the cost of increased hydraulic resistance. Studies of mechanical vibration and flow oscillation demonstrate notable enhancements in heat transfer in low-Reynolds-number and buoyancy-dominated regimes when vibration parameters are optimally tuned. The integration of porous media, including metal foams and packed spheres, has shown substantial performance gains, often exceeding 40–90%, though significant pressure-drop challenges accompany this approach. More recently, artificial intelligence and data-driven optimization techniques have emerged as powerful tools for balancing thermal enhancement and hydraulic penalties. Despite these advances, key gaps persist in condensation-dominated applications, low-Reynolds-number regimes, long-term reliability, and experimentally validated coupled thermal–hydraulic optimization. This review consolidates existing knowledge, identifies unresolved challenges, and outlines future research directions towards high-efficiency, application-specific cross-flow heat exchanger design.

Keywords: Crossflow heat exchanger; Compact heat exchanger; Thermal-hydraulic performance; Vortex generator; Flow-induced vibration

1 Introduction

Heat exchangers operate on the fundamental principle of transferring thermal energy between two or more fluids at different temperatures without direct mixing. This process is governed by the laws of energy conservation and by heat conduction and convection mechanisms. Heat flows from the higher-temperature fluid to the lower-temperature fluid across a solid separating wall or through direct contact in specialized configurations. The rate of heat transfer depends on the overall heat-transfer coefficient, the temperature driving force, the heat-transfer area, and the fluid’s thermophysical properties. Various flow arrangements—such as parallel, counterflow, and crossflow—affect the temperature distribution and thermal effectiveness. Counterflow arrangements typically provide the highest effectiveness due to the more uniform temperature gradient. Heat exchangers are designed to balance thermal performance, pressure-drop constraints, mechanical durability, and fouling resistance. Their operation is vital to thermal systems for activities such as heating, cooling, condensation, evaporation, and energy recovery in various industrial contexts [1]. Numerous types of heat exchangers are utilized across industrial and scientific applications, each differing in construction, flow arrangement, and heat-transfer characteristics. These categories, along with their distinguishing features, are systematically classified and illustrated in Figure 1 [2, 3].

Crossflow heat exchangers are manufactured in a wide range of geometries and structural configurations to accommodate the thermal and operational demands of different applications. These variations include differences in

fin design, flow arrangement, material selection, and compactness. A comprehensive overview of the specifications and distinguishing features of each crossflow configuration is presented in Table 1.

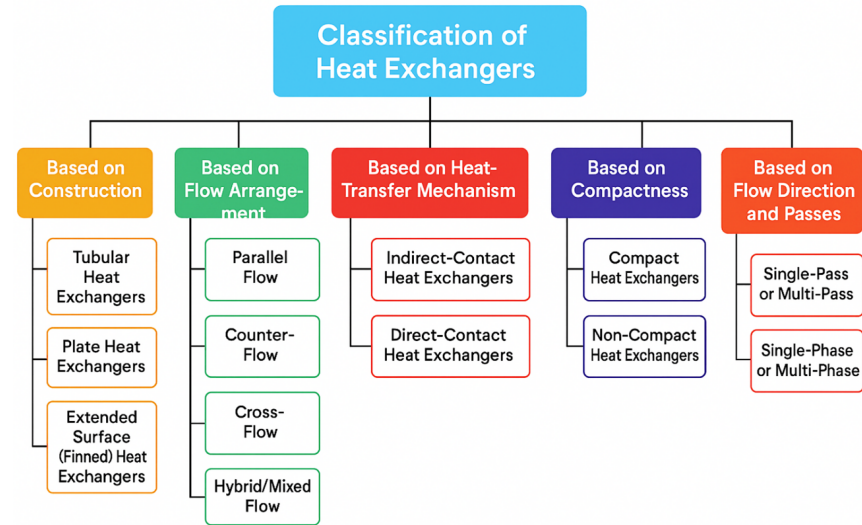


Figure 1. A detailed illustration of the different classifications of heat exchangers

Table 1. Comprehensive details of crossflow heat exchanger classifications [4, 5]

Heat Exchanger Type	Geometry	Typical Applications	Key Features
Plate-fin crossflow heat exchanger	Stacked plates with offset, wavy, or louvered fins	Automotive radiators, aerospace cooling, gas-gas exchangers	High surface-area density; compact; lightweight; enhanced turbulence
Tube-fin crossflow heat exchanger	Circular/oval tubes arranged transversely with continuous or segmented fins	Air conditioning units, refrigeration	Robust construction; suitable for air-side heat transfer; increased surface area
Microchannel crossflow heat exchanger	Parallel microchannels with flat tubes and microfins	Industrial cooling, chemical processing	High heat-transfer coefficient; low refrigerant charge; compact size
Crossflow plate heat exchanger (Plain)	Corrugated/flat plates forming orthogonal flow passages	Energy recovery ventilators	High effectiveness; suitable for gas-gas heat exchange
Crossflow regenerative heat exchanger	Rotary wheel or fixed matrix alternately exposed to hot and cold streams	Energy recovery ventilators, power plants	High efficiency; moisture/heat recovery; periodic flow reversal
Extended surface heat exchanger (Finned crossflow)	Tubes or plates with extended fins on the air side	Air coolers, condensers, radiators	Enhanced air-side heat transfer; good performance with low-conductivity fluids

2 Review Methodology

This review provides a structured, comprehensive assessment of cross-flow heat exchanger technologies, covering both classical developments and recent advances in heat transfer enhancement techniques. To ensure transparency and reproducibility, a systematic literature review was conducted. A comprehensive search was conducted across major scientific databases, including Scopus, Web of Science, ScienceDirect, and Google Scholar. The search strategy combined keywords such as “cross-flow heat exchanger,” “heat transfer enhancement,” “fin design,” “vortex generator,” “porous media,” “nanofluid,” “optimization,” “machine learning,” and “thermo-hydraulic performance.” Boolean operators (AND, OR) were used to refine the search and ensure broad coverage. The selection of relevant studies followed predefined criteria:

- **Relevance:** Studies focused on cross-flow heat exchangers and enhancement techniques
- **Quality:** Peer-reviewed journal articles and high-quality conference papers
- **Time span:** Inclusion of both foundational studies and recent publications, with an emphasis on post-2015 developments
- **Data availability:** Studies reporting quantitative performance indicators (e.g., Nusselt number, pressure drop, performance evaluation criterion (PEC), or j/f ratio)

The literature screening process was conducted in multiple stages, including identification, screening, eligibility assessment, and final inclusion. Duplicate records were removed, and titles and abstracts were evaluated to exclude irrelevant studies. Full-text articles were then assessed against the defined criteria. The complete selection process is illustrated in Figure 2, following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework.

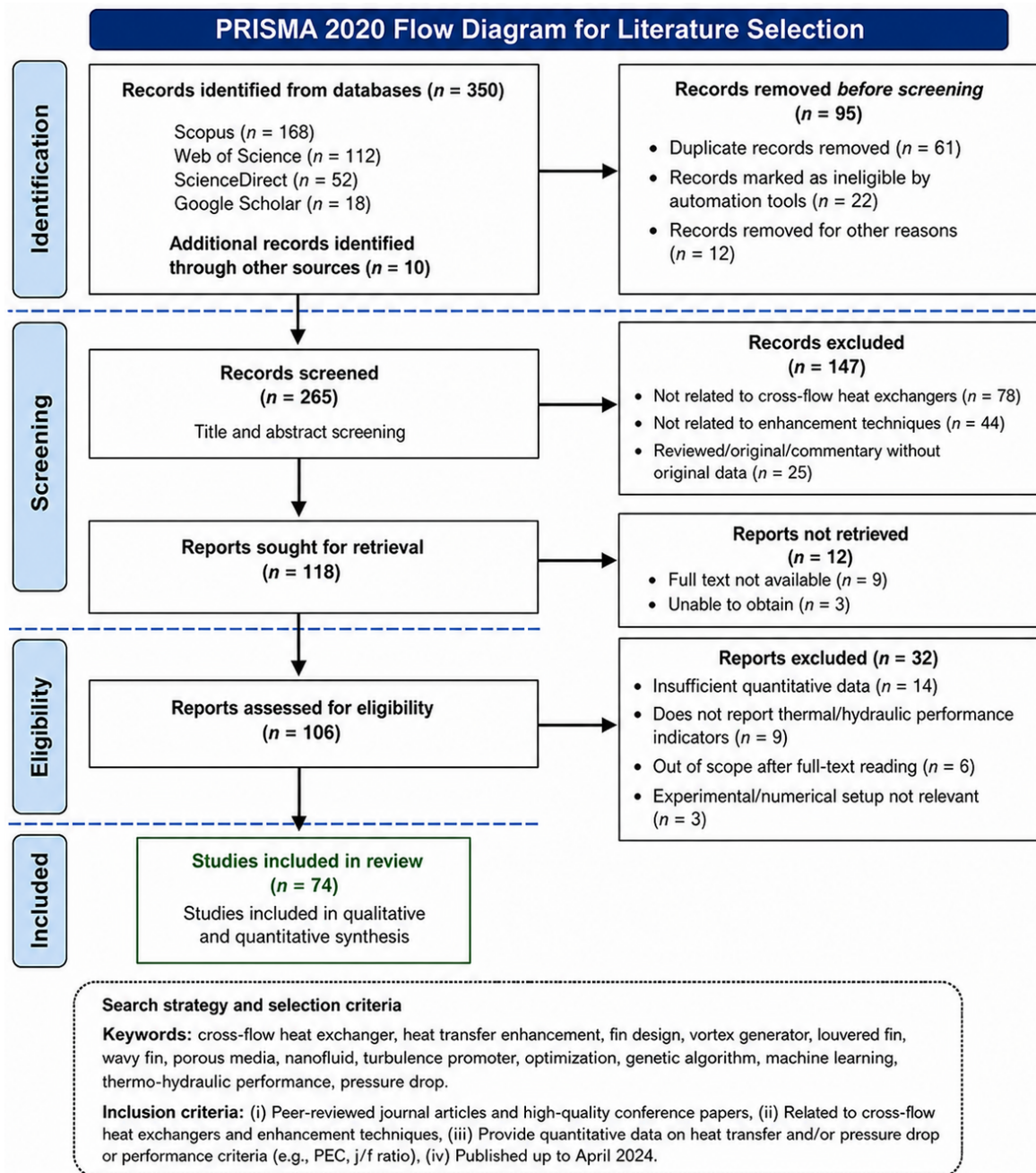


Figure 2. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram illustrating the identification, screening, eligibility, and inclusion process of the reviewed literature

3 Classification of the Enhancement Techniques

Crossflow heat exchangers play a vital role in a wide range of industrial and service applications, including power generation, Heating, Ventilation, and Air Conditioning (HVAC) systems, refrigeration units, and chemical processing operations. Their significance arises from their compact structure, high surface-area density, and ability to accommodate diverse operating conditions. As a result, continuous efforts have been devoted to enhancing their thermohydraulic performance. Over the past four decades, extensive research has been reported in the form of theoretical analyses, advanced numerical modeling, and controlled experimental investigations. These studies focus on improving heat-transfer effectiveness, reducing air-side resistance, optimizing fin and tube geometries, and applying extended or enhanced surfaces to augment performance. Moreover, advancements in computational fluid dynamics (CFD) and experimental diagnostics have enabled more profound insights into flow behavior, turbulence patterns, and local heat-transfer distributions. In the following segment, we compile essential conclusions from selected studies conducted over the past 40 years, focusing on key advancements, technological enhancements, and strategies to improve the performance of crossflow heat exchangers. These are classified into the following categories:

3.1 Geometric Modification Techniques

Žukauskas and Ulinskas [6] developed empirical correlations to evaluate heat transfer and pressure drop in tube banks exposed to crossflow. Analyzing extensive experimental data across wide Reynolds and Prandtl number (Pr) ranges, they proposed efficiency parameters that accounted for geometric variations, including transverse and longitudinal pitches. The study provided practical correlations for predicting Nusselt number and friction factor, supporting optimized tube-bank design for improved thermal performance and flow resistance management.

Molki and Yuen [7] investigated the influence of intwall spacing on convective heat transfer and pressure drop in parallel-plate flow. Using experimental measurements and flow-visualization techniques, the study varied gap spacing and flow conditions to characterize how spacing affects the local flow structure, heat-transfer coefficient, and pressure drop. The results showed that reducing the inter-wall spacing enhances the heat-transfer coefficient due to increased turbulence and mixing, but also significantly increases the pressure drop. These findings provide critical guidance for the geometric optimization of compact heat-exchange passages where tight spacing can improve thermal performance but at the expense of hydraulic loss.

Žukauskas [8] investigated convective heat transfer from tubes in crossflow, aiming to develop general correlations applicable to a range of flow and thermal conditions. Using experimental data, the study examined heat transfer coefficients for various tube arrangements, including inline and staggered layouts. The work produced empirical correlations widely used in heat exchanger design, accounting for tube geometry, Reynolds number (Re), and Pr . This study remains a seminal reference for crossflow tube bundle analysis.

Monheit and Freim [9] studied the impact of tube-bank inclination on the thermal-hydraulic performance of air-cooled heat exchangers. Using experimental methods, they compared the performance of inclined staggered tube banks with that of standard horizontal configurations. Variables such as face velocity, heat transfer coefficient, and pressure drop were analyzed. Their findings showed that inclination alters both heat transfer and pressure characteristics, leading to the development of correction factors for performance prediction. This work provides practical insights for designing heat exchangers with geometric or installation constraints.

Jensen et al. [10] examined the impact of tube bundle geometry on boiling heat transfer and pressure drop during crossflow. The study aimed to determine how tube arrangement parameters—such as pitch, layout, and diameter—affect phase-change heat transfer and fluid resistance. Utilizing experimental methods, the authors conducted tests under various boiling conditions and geometric configurations. The results indicated that tube spacing and arrangement significantly influence heat transfer coefficients and pressure drop characteristics. The findings underscore the importance of tube bundle geometry optimization to enhance thermal performance while minimizing flow resistance in crossflow boiling systems.

Achenbach [11] investigated heat transfer from a staggered tube bundle in crossflow at high Re using experimental measurements of local and overall Nusselt numbers and pressure drop. The study examined a seven-row, three-column staggered configuration with pitch ratio $a = 2.0$ under smooth and rough tube surface conditions. Key findings revealed that increasing the Re significantly enhances heat transfer coefficients while increasing pressure drop; the rough-surface bundle exhibited higher heat transfer at the cost of increased hydraulic resistance.

Nir [12] developed empirical correlations for heat transfer and friction factor in crossflow over staggered finned-tube banks. Using experimental data from various geometries with plain and segmented fins, correlations were derived as functions of Re , fin and tube spacing, and geometry. The proposed correlations predict heat-transfer coefficients and friction factors with approximately $\pm 20\%$ accuracy. This study provides practical design tools for engineers optimizing finned tube-bank heat exchangers under crossflow conditions.

Manglik and Bergles [13] developed empirical correlations for heat transfer (expressed as the Nusselt number) and pressure drop (expressed via the friction factor) in rectangular offset-strip-fin compact heat exchangers. Their experimental study varied fin pitch, strip height, Re , and fin geometry to capture performance trends. The correlations

enable the prediction of thermal and hydraulic behavior across a wide range of operating conditions and geometries, offering practical tools for compact heat exchanger design and optimization.

Wang et al. [14] conducted a study on the convective heat transfer and friction characteristics of wavy fin-and-tube heat exchangers under crossflow conditions. In their experimental research, they varied key geometric parameters such as fin pitch, wave amplitude, and tube arrangement to assess how these factors influence the Nusselt number and friction factor based on flow conditions and geometry. The findings showed that wavy fin configurations significantly improve heat transfer compared to plain fins, though this is accompanied by increased pressure drop. The correlations developed in this study provide valuable design tools for optimizing the thermal performance of wavy fin-and-tube heat exchangers.

Martin et al. [15] conducted a numerical study of the laminar crossflow around periodic, sparsely arranged cylinder arrays to assess frictional losses and convective thermal performance. Employing CFD simulations on unit cells of the array, they varied the Re and porosity to evaluate pressure drop and Nusselt number behavior. The findings revealed that increased spacing (higher porosity) reduces pressure drop but also diminishes convective heat transfer; the study offers correlations linking geometry, flow conditions, and performance for the design of periodic tube-bank systems in crossflow.

Lange et al. [16] explored momentum and heat transfer from cylinders in laminar crossflow within a Re range of $10^{-4} \leq Re \leq 200$. Using two-dimensional numerical simulations of flow around a heated circular cylinder, they quantified the behavior of the Nusselt and drag coefficients as functions of Re and the Pr . Results revealed that both heat-transfer rates and drag forces exhibit distinct changes as Re crosses key thresholds, providing unified correlations for laminar crossflow around cylinders.

Li and Kottke [17] conducted a study to visualize and determine local heat transfer coefficients in shell-and-tube heat exchangers with staggered tube arrangements. They employed mass transfer measurements as part of their experimental methodology to map the distributions of local Sherwood numbers, which were then converted to Nusselt numbers for the tube outer surfaces. Their findings revealed significant spatial variations in heat transfer characteristics due to wake interactions and bypass flows, offering detailed insights into the thermal behavior on the shell side. These results can help improve the design and optimization of staggered-tube shell-and-tube heat exchangers.

Ranganayakulu and Seetharamu [18] conducted a numerical study on crossflow plate-fin heat exchangers to analyze the combined effects of wall longitudinal heat conduction, flow maldistribution, and temperature nonuniformity on thermal performance. Their finite-element model incorporated two-dimensional conduction in the fins and plates, accounted for flow nonuniformity in both inlet streams, and addressed nonuniform temperature distributions. The results indicated that these non-uniformities and longitudinal conduction work together to reduce heat exchanger effectiveness by as much as 15%. The authors recommended design adjustments to help mitigate these performance losses.

Yu and Tao [19] conducted experiments on turbulent flow within annular tubes equipped with internal, longitudinal wave-shaped fins. They measured the Nusselt number and friction factor for Re ranging from 900 to 3,500, comparing tubes with both blocked and unblocked inner inserts. Their findings demonstrated a significant enhancement in heat transfer, particularly with the use of blocked-core inserts, though this also increased the friction factor. The study provided correlations for the Nusselt number as approximately $Nu \approx Re^{0.8}$ and for the friction factor as approximately $f \approx Re^{-0.4}$ in the developed flow regime.

Erek et al. [20] conducted a study to investigate how geometric parameters affect air-side heat transfer and pressure drop in a plate-fin-and-tube heat exchanger using numerical simulations. They varied several factors, including fin pitch, tube ellipticity, and fin height, across a range of Re (100–1,000). The results indicated that increasing tube ellipticity improves heat transfer while simultaneously reducing pressure drop. In contrast, using a tighter fin pitch significantly increases pressure losses with only a slight improvement in heat transfer. These findings provide valuable design guidance for optimizing the geometry of fin-tube exchangers.

Junqi et al. [21] developed empirical correlations to predict heat transfer and pressure drop in compact crossflow heat exchangers that use wavy fins and flat tubes. Their experiments varied fin pitch, tube layout, and air velocity across Re of up to approximately 3,000. The results showed that wavy-fin configurations improved heat transfer by up to 28% compared to plain fins, but also increased pressure drop by 20–25%. These findings highlight the importance of balancing enhanced heat transfer with acceptable hydraulic performance in the design of fin-and-tube heat exchangers.

In 2007, Dong et al. [22] developed empirical correlations for heat transfer and pressure drop in compact heat exchangers that feature multi-louvered fins. They conducted extensive experiments on 20 different configurations with variations in fin pitch, louver angle, and air velocity ($Re = 400$ –3,000). Their findings revealed that optimizing louver geometry significantly enhances heat transfer, increasing the Colburn J -factor by up to 40%. However, this improvement is accompanied by a 25% increase in the friction factor. The resulting correlations provide accurate performance predictions for the design of louvered-fin exchangers.

Ibrahim and Gomaa [23] studied the thermal-hydraulic performance of elliptic tube bundles in crossflow, focusing on key criteria such as the Nusselt number, pressure drop, and “goodness factor.” They conducted a three-dimensional numerical analysis, varying the axis ratio, angle of attack, and longitudinal/transverse pitches over a Re range of approximately 5,500 to 14,500 in staggered arrangements. Their results indicated that bundles with lower axis ratios, smaller angles of attack, and reduced longitudinal spacings achieved a significantly higher goodness factor—up to 8 times that of circular tubes, allowing for more compact and efficient heat exchanger designs.

Durmuş et al. [24] conducted a study on the air-side heat transfer and pressure drop characteristics of plate heat exchangers featuring three different surface profiles: flat, corrugated, and an alternative geometry, all under single-phase flow conditions. They performed experimental measurements over a range of Re to assess how the Nusselt number and friction factor depend on surface geometry. The results indicate that corrugated profiles enhance heat transfer by approximately 30% compared to flat plates, though they also result in higher pressure drops. This study highlights the significance of optimizing surface profiles for the efficient design of plate heat exchangers.

Dović et al. [25] developed generalized correlations to predict the Nusselt number and friction factor in plate heat exchanger channels with arbitrary geometries. By modeling a representative channel cell and calibrating the model using extensive experimental data, the study formulated expressions applicable to both laminar and turbulent flow regimes. The correlations achieved average absolute errors of approximately 4% for the friction factor and 5% for the Colburn j -factor across corrugation angles ranging from 30° to 80° . These results significantly enhance the accuracy of preliminary designs for plate heat exchangers.

Lee and Kim [26] conducted a numerical investigation into the thermal-hydraulic performance of zigzag printed circuit heat exchangers (PCHEs) using three-dimensional conjugate heat transfer simulations. They analyzed four channel shapes—semicircular, rectangular, trapezoidal, and circular—and various zigzag configurations across a broad range of Re . The results indicate that rectangular channels achieve the highest Nusselt number and effectiveness, but also exhibit the highest friction factor. In contrast, circular channels demonstrate the weakest thermal performance. Additionally, both the Nusselt number and effectiveness increase with the Re , while the j -factor and friction factor decrease.

Liang et al. [27] conducted combined experimental and simulation analyses to evaluate the air-side thermal-hydraulic performance of automotive heat exchangers under varying flow rates and fin geometries. They employed distributed parameter modeling and the number of transfer units (NTU)-effectiveness approach to measure heat transfer coefficients, pressure drops, and the effects of flow mal-distribution. The results showed strong agreement between the simulations and experiments, enabling the development of correlations for the Nusselt number and friction factor. These findings will aid in the improved design of compact air-side heat exchanger modules for automotive applications.

Khaled et al. [28] explored the feasibility and thermal performance of using a multi-pass design in water–air crossflow tube-and-fin heat exchangers. They developed a detailed thermal model based on the ε -NTU methodology, combined with geometric and operational constraints, to assess how dividing the air passage into multiple sequential sections impacts thermal effectiveness and pressure drop. The results indicated that the multi-pass configuration improved thermal effectiveness by 12–20%, depending on the water-side Re and fin spacing, while the associated pressure drop remained moderate, at less than 15%. This study demonstrates that multi-pass channeling can significantly enhance the compactness and thermal performance of heat exchangers without incurring substantial hydraulic drawbacks.

Zhang et al. [29] investigated the thermal-hydraulic performance of a circular tube-and-fin heat exchanger enhanced using ellipsoidal protrusions embossed on the fin surfaces. Through combined experimental testing and CFD simulation, the study evaluated how protrusion size, spacing, and orientation influence air-side heat transfer. Results showed that ellipsoidal protrusions increased the Colburn j -factor by 14–32%, depending on geometry, while the friction factor increased by 10–28%. The improvement was attributed to strengthened flow mixing and disruption of boundary-layer development around the fins. Overall, the protruded-fin configuration achieved higher $PEC (>1)$, confirming its potential as a passive enhancement technique for compact air-side heat exchangers.

Wu et al. [30] conducted an experimental study to assess the thermal performance of a finned-tube heat exchanger under frosting conditions. They tested both flat and corrugated fin geometries in a controlled environment with high humidity and sub-zero airflow to measure the impact of frost on performance. The results indicated that frosting reduced air-side heat transfer by 18–35% and increased pressure drop by 25–50%. Notably, the corrugated fins experienced the most significant deterioration due to thicker frost accumulation. In contrast, flat fins exhibited 10–15% greater thermal stability than corrugated fins, highlighting the importance of selecting the appropriate fin geometry for refrigeration and heat pump systems operating in frost-prone environments.

Rukruang et al. [31] experimentally investigated the performance of a novel heat exchanger with an effective cross-section flattened (ECF) tube to understand its behavior under single-phase flow. The analysis was conducted for the alternating cross-section flattened (ACF) tubes and compared with that for conventional circular tubes under various flow conditions. The results showed that the ACF design not only achieved a higher Nusselt number but

also a larger overall heat transfer coefficient, with an acceptable pressure drop. The performance evaluation using j - and f -factors indicates that ACF tubes exhibit superior heat transfer characteristics with an acceptable pressure drop; therefore, they are suitable for use in compact heat exchangers.

Lee et al. [32] conducted both experimental and numerical evaluations of the air-side performance of wavy fin-and-tube heat exchangers that use elliptic tubes with large waffle-height fins. In their study, they utilized a four-row configuration to measure heat transfer and pressure drop over a range of inlet air velocities. The results indicated that increasing waffle height increased the Colburn j -factor by 12–28%, while the friction factor rose by 10–22%, depending on fin pitch and wave geometry. Additionally, the study found that Wang's correlation provided the most accurate predictions for the j and f values for this geometry. These findings underscore the sensitivity of wavy-fin elliptic-tube exchangers to factors such as waffle height, fin pitch, and wave structure.

Song et al. [33] proposed a novel fin configuration featuring ellipsoidal dimple protrusions for a circular-tube fin heat exchanger and numerically investigated its thermo-hydraulic performance over the range $Re = 1,500$ – $5,000$. The design significantly enhanced secondary flow and heat transfer, yielding a Nusselt number up to 29% higher than in a smooth channel and outperforming conventional vortex generators (VG). The maximum thermal performance factor reached 1.161, with optimal performance at $\beta \approx 20^\circ$, balancing heat transfer and pressure drop.

Maghsoudali et al. [34] conducted an experimental study to evaluate the thermal and hydraulic performance of helix-wire finned-tube heat exchangers. The research focused on the influence of helix pitch, wire diameter, and fin height across a wide range of Re . The results indicated that helix-wire fins improved the heat transfer coefficient by 18–42% compared to plain tubes, while the friction factor increased by 15–33%, depending on the geometry. The best overall performance was observed at intermediate helix pitches, where the swirl-induced secondary flow enhanced convective mixing without causing excessive pressure drops. These findings support the use of helix-wire fin designs as a promising passive enhancement technique for compact heat exchanger applications.

Liu et al. [35] conducted both experimental and numerical studies on internally finned tubes with variable cross-sections, analyzing their performance over a Re range of $5,000$ – $25,000$. The gradually varying fin geometry increased swirl and secondary flows, leading to Nusselt numbers that were 22% to 48% higher than those of constant-section fins. Additionally, the friction factor increased by 18% to 40% with variations in fin height. The CFD predictions aligned with the experimental results within 5%, demonstrating their reliability. Furthermore, new correlations for the Nusselt number and the friction factor were developed to aid in the design of enhanced internal cooling passages.

Studies show that changing fin and tube designs can improve heat transfer in crossflow heat exchangers. Techniques such as wavy fins, louvered fins, elliptical tubes, and surface protrusions boost thermal performance by disrupting boundary layers and promoting flow mixing, but often increase pressure drop, especially with obstructive designs. The results of these studies show that some designs (e.g., elliptical tubes, protrusions) achieve high PEC (>1), whereas others yield limited gains due to hydraulic penalties. Many studies focus only on heat transfer without optimizing for pressure loss. Geometric modifications work best where compactness and high heat transfer are needed, and moderate pressure drops are acceptable, such as in automotive radiators and air heat exchangers.

3.2 Flow Control Techniques

Obot and Trabold [36] experimentally investigated heat transfer in circular jet arrays under minimum, intermediate, and complete crossflow conditions, considering both small and large jet-to-target spacings. Parameters such as the jet Re ($1,000$ – $21,000$), the jet-to-surface spacing (2 – 16 diameters), and the open area ratio were varied. Results showed that minimum crossflow yielded the highest heat-transfer performance, while increased crossflow degraded effectiveness. Additionally, increasing the jet number at fixed blower power improved heat transfer, especially at narrow spacing under minimum crossflow.

Ranganayakulu et al. [37] experimentally investigated the impact of inlet flow maldistribution on the thermal and hydraulic performance of crossflow plate-fin compact heat exchangers. The study analyzed how nonuniform inlet conditions affect heat-transfer effectiveness and pressure drop. Results showed that maldistribution significantly reduces performance while increasing pressure losses. The authors proposed correction factors to account for inlet nonuniformity, offering valuable guidelines for improving the design and evaluation of compact heat exchangers operating under realistic, non-ideal flow conditions.

Dutta and Dutta [38] experimentally investigated how baffle size, perforation pattern, and orientation influence internal heat-transfer enhancement in channel flows. By varying baffle geometry and flow parameters, they measured changes in Nusselt number and friction factor under turbulent conditions. The results showed that optimally sized, perforated baffles oriented at specific angles yield significant heat-transfer benefits, though at the cost of increased pressure drop. Their findings provide practical design correlations for internal thermal-hydraulic optimization of baffled channels.

In 2003, Tiwari et al. [39] used oval tubes and multiple delta-winglet VG. By adjusting the winglet angles and positions, the study demonstrated that heat transfer could be increased by up to 35% compared to circular-tube designs, particularly in the wake zones. While there was a moderate increase in pressure drop, this indicated a

trade-off between thermal performance and hydraulic resistance. This configuration offers an effective strategy to enhance the efficiency of compact heat exchangers.

Joardar and Jacobi [40] investigated the effects of leading-edge delta-wing VG on a flat-tube, louvered-fin compact heat exchanger in a crossflow configuration. Their full-scale wind tunnel experiments, conducted on automotive-type units under both dry and wet conditions, revealed that the addition of delta-wing VG improved average heat transfer by 21% in dry conditions and 23.4% in wet conditions, while resulting in a pressure drop of less than 7%. These findings underscore the potential of using VG to enhance the design of compact heat exchangers.

Joardar and Jacobi [41] conducted an experimental study on winglet-type VG arrays integrated into a compact plain-fin and tube heat exchanger, specifically under air-side crossflow conditions with Re ranging from 220 to 960. They found that a single-row VG arrangement increased the air-side heat transfer coefficient by 16.5% to 44%, while incurring a pressure drop of less than 12%. In contrast, a three-row VG array achieved greater enhancements, with a heat transfer coefficient increase of 29.9% to 68.8%, but resulted in significantly higher pressure drops (ranging from approximately 26% at $Re = 960$ to around 87.5% at $Re = 220$). These results emphasize the trade-off between improved heat transfer and increased hydraulic costs in the design of compact heat exchangers.

Wang et al. [42] conducted experiments to evaluate heat transfer enhancement on the shell side of a shell-and-tube heat exchanger. They installed sealing strips to reduce bypass flow and measured improvements in the shell-side heat transfer coefficient ranging from 18.2% to 25.5%. Additionally, the overall heat transfer increased by up to 17%. These results demonstrate that flow-control modifications can significantly improve the heat exchanger's performance. The findings provide practical design strategies for improving shell-side performance in industrial heat exchangers.

Thianpong et al. [43] conducted an experimental evaluation of the thermal performance of circular tubes equipped with twisted-ring turbulators at Re ranging from approximately 6,000 to 20,000 using air as the working fluid. They tested three width ratios ($W/D = 0.05\text{--}0.15$) and three pitch ratios ($P/D = 1\text{--}2$). The results showed that increasing the width ratio improved both the Nusselt number and the friction factor, whereas higher pitch ratios reduced them. The optimal thermal performance factor, which was about 1.24, was achieved with the smallest width and pitch ratios. Additionally, the study developed empirical correlations for both the Nusselt number and the friction factor.

Hanuszkiewicz-Drapa la et al. [44] employed a numerical model to examine the thermal-hydraulic performance of cross-flow heat exchangers under real operational conditions, with particular emphasis on uneven media flow distribution. The model encapsulates flow maldistribution by incorporating non-uniformities between channels and non-uniform inlet and outlet boundary conditions. This approach facilitates the assessment of the reduction in operational effectiveness caused by non-uniform flow. The authors concluded that flow profile distortion can reduce heat transfer efficiency by up to 20%, indicating a substantial impact of flow maldistribution on heat exchanger performance. The study offers valuable insights into the design and diagnostics of cross-flow heat exchangers for practical applications.

Li et al. [45] conducted an experimental study to investigate the impact of mechanical vibration on heat transfer in a fin-tube vehicle radiator under standard driving conditions. Using the ε -NTU method, they found that applying vibration increased the air-side Nusselt number by 2.98% to 16.82%. The study demonstrated that forced vibration enhances convective heat dissipation in automotive radiators, suggesting a potential way to improve thermal performance in vehicle cooling systems.

Fu et al. [46] investigated heat transfer in a rectangular channel under mechanical vibration using both numerical and experimental methods. Their study used field-interaction analysis and validated CFD modeling to examine the effects of vibration amplitude and frequency on thermal-hydraulic performance. They found that a 2 mm vibration amplitude at 25 Hz increased the average Nusselt number by 18–27%, while the friction factor rose modestly by 8–12%. Beyond a 4 mm amplitude, heat transfer improvement plateaued, and hydraulic resistance increased sharply by up to 35%. This performance enhancement resulted from improved alignment between vibration-induced velocity gradients and local temperature gradients. The authors also proposed performance indices, including the j/f ratio and the Thermal-Vibration Performance Factor (TVPF), to quantify benefits across different vibration levels.

Song et al. [47] conducted a study on the thermal and hydraulic characteristics of fin-and-tube heat exchangers that are equipped with wavy delta-winglet VG. They used both experiments and CFD across a range of air-side Re from 500 to 2500. The wavy VG generated strong longitudinal vortices, enhancing mixing and reducing the boundary layer thickness. As a result, heat transfer was improved by 18% to 32%, while the pressure drop increased by 20% to 45%, depending on the fin pitch and wave amplitude. Overall, the thermal performance improved moderately, highlighting the advantages of wavy VG for compact air-side heat exchangers.

Zhao et al. [48] studied the interaction between cross-flow-induced vibration (FIV) and convection heat transfer in tube bundles operating at subcritical Re (5×10^3 to 3×10^4). Through validated 3-D fluid-structure interaction simulations, they examined turbulence-induced vibration, vortex-induced resonance, and fluid-elastic instability. The results indicated that vortex-induced resonance increased the average Nusselt number by approximately 8.8%. In contrast, fluid-elastic instability led to a higher pressure drop with only limited thermal benefits. At low flow

velocities, the vibration's impact was negligible. This study clarifies the conditions under which FIV can either enhance or degrade the performance of tube bundles.

Wang et al. [49] investigated the effect of vertical vibration (representative of marine operating conditions) on the thermal-hydraulic performance of a brazed plate heat exchanger (BPHE). The study combined experimental testing with detailed measurements of heat transfer and pressure drop across varying vibration intensities. Results showed that under vibration, the BPHE's heat transfer coefficient increased by up to 13.6%. At the same time, pressure drop remained within acceptable limits, demonstrating that vibration can enhance convective performance in compact heat exchangers without a severe hydraulic penalty.

Kaushik et al. [50] conducted a numerical investigation into the effects of inclined splitter plates on flow control, wake dynamics, heat transfer, and pressure drop in cross-flow heat exchangers featuring circular tubes. They assessed various splitter-plate length-to-diameter ratios ($L/D = 0, 0.5, 2, 3$) and inclination angles ($\alpha = 0^\circ, 15^\circ, 30^\circ$) at Re ranging from 5,500 to 14,500. The results revealed a maximum Nusselt number enhancement of 78.8% at an inclination angle of 30° and a Re of 5,500. For Re of 11,500 and 14,500, Nusselt numbers increased by approximately 50.9% and 47.6%, respectively. Among the various geometric configurations tested, the $L/D = 2$ ratio demonstrated the best overall performance, yielding the highest PEC before the friction penalties became excessive. These findings indicate that inclined splitter plates are an effective passive strategy for significantly improving the performance of cross-flow heat exchangers by manipulating the flow field.

Flow control techniques such as VG, baffles, jet impingement, and vibration enhance heat transfer by modifying the flow and inducing secondary flows. They disrupt boundary layers and boost turbulence, improving performance. VG and splitter plates often achieve $PEC > 1$, indicating good thermo-hydraulic performance. However, multi-row VG or strong obstructions can cause high pressure drops, reducing efficiency. Vibration methods work well at low Re but require tuning to prevent performance loss. Limitations include dependence on operating conditions and system complexity, making these techniques suitable mainly for applications like compact heat exchangers in HVAC and aerospace.

3.3 Advance Material and Prous Media

Dai and Sumathy [51] conducted a theoretical analysis of a crossflow direct evaporative cooler that utilizes honeycomb paper as the packing material. Their mathematical model included heat and mass transfer balances for humid air flowing through the wetted paper channels. The parameters studied comprised air and water flow rates, channel dimensions, and pad thickness. The results indicated that optimal cooling performance was achieved at specific channel lengths and air-to-water mass flow ratios, with temperature reductions of up to 15°C possible under arid conditions. This study provides valuable design guidelines for passive cooling applications.

Zhan et al. [52] conducted numerical analyses of an indirect evaporative cooler that utilizes an M-cycle crossflow heat exchanger. Their finite-element model addressed the coupled heat and mass transfer for both dry and wet air streams. The parametric results indicated that lower channel air velocity, reduced inlet humidity, and a working-to-product air mass flow ratio of approximately 50% can maximize cooling effectiveness. Under optimal conditions, the M-cycle configuration demonstrated up to 16.7% greater cooling effectiveness compared to conventional crossflow devices.

Monteiro and de Mello [53] conducted CFD simulations to evaluate the thermal performance and pressure drop of a ceramic plate-fin heat exchanger designed for high-temperature applications. Their simulations, conducted for Re ranging from 500 to 1500, enabled them to develop correlations for the Colburn j -factor and friction factor, with prediction errors of $\pm 5\%$ or less. The results indicated that the ceramic core maintained stable thermal performance up to 800°C , while increases in pressure drop remained within acceptable limits. This study confirms that ceramic heat exchangers are reliable candidates for high-temperature crossflow thermal systems.

Peyghambarzadeh et al. [54] conducted an experimental study on dilute nanofluids, specifically CuO and Fe_2O_3 suspended in water at concentrations ranging from 0.15% to 0.65% by volume, using a car radiator setup. They tested the fluids at liquid-side Re between 50 and 1000 and inlet temperatures from 50°C to 80°C . The results showed that both nanofluids increased the overall heat transfer coefficient by up to approximately 9% compared to water. They observed that higher nanoparticle concentrations and higher fluid velocities increased the heat transfer coefficient (U), whereas higher inlet temperatures decreased it.

Ray et al. [55] conducted an experimental study on three types of water-based nanofluids—suspensions of Al_2O_3 , CuO , and SiO_2 nanoparticles—in an automotive radiator under conditions similar to engine coolant flow. They found that using these nanofluids increased heat transfer rates and reduced coolant exit temperatures compared to the base fluid. Among the nanofluids tested, the Al_2O_3 suspension showed the best performance. Additionally, the associated pressure drop and pumping power losses were modest. These findings suggest that nanofluids can significantly enhance the cooling performance of vehicle radiators.

Selvam et al. [56] conducted experiments on a car radiator using a mixture of water and ethylene glycol (70:30), which was seeded with graphene nanoplatelets at concentrations of 0.1% to 0.5% by volume. The mass flow rates

varied from 10 to 100 g/s, and the inlet temperatures were set at 35 °C and 45 °C, with a fixed air velocity of 3 m/s. The results showed that the convective heat transfer coefficient increased with higher nanoplatelet loading, inlet temperature, and flow rate. Specifically, at a concentration of 0.5% by volume and a flow rate of 100 g/s, enhancements of approximately 20% at 35 °C and around 51% at 45 °C were observed. Additionally, the pressure drop also increased with greater nanoplatelet loading and higher flow rates.

The integration of advanced materials, including porous media, nanofluids, and high-temperature ceramics, is a powerful strategy for enhancing heat exchanger performance. Porous structures significantly increase the effective surface area and promote mixing, yielding substantial heat transfer enhancement—often 40–90%. Similarly, nanofluids improve thermal conductivity and convective heat transfer, particularly on the liquid side. However, a major limitation of these approaches is the associated increase in pressure drop, especially in porous media, where flow resistance can dominate, leading to PEC values below unity. Additional challenges include nanofluid stability, material cost, and long-term durability under operational conditions. Despite these limitations, these techniques are highly suitable for applications requiring maximum heat transfer enhancement in constrained volumes, such as electronic cooling, high-temperature systems, and compact energy devices, provided that pressure-drop penalties can be managed.

3.4 Optimization and Data-Driven Methods

Sekulić and Herman [57] proposed a thermodynamic approach to minimize irreversibility in the design of compact crossflow heat exchangers. Using second-law analysis, they developed a theoretical model that evaluates entropy generation as a function of geometry, heat capacity rate ratio, and flow arrangement. The study demonstrated that reducing local entropy generation increases thermal efficiency and improves overall performance. Their work introduced entropy generation minimization (EGM) as a fundamental design strategy, establishing a systematic framework for optimizing compact heat exchangers beyond conventional first-law performance criteria.

Ravikumar et al. [58] performed a finite element analysis to investigate the thermal performance of crossflow compact heat exchangers. The study focused on evaluating the influence of key parameters, such as the Re and heat capacity rate ratio, on heat exchanger effectiveness. By solving the governing energy equations using the finite element method (FEM), they modeled different geometrical configurations. The results demonstrated that effectiveness improves with optimized capacity ratios and geometry, establishing FEM as a robust tool for the analysis and design of compact heat exchangers.

Shah and Pignotti [59] performed a thermal analysis of complex crossflow heat exchangers by relating them to standard single-pass configurations. Their objective was to simplify the performance prediction of multi-pass or multi-row crossflow devices by mapping them onto equivalent known forms. They used a matrix-based method and the effectiveness-NTU approach to evaluate cases with up to ten passes. Findings revealed that beyond an optimal NTU value, additional surface area yields minimal gains, thus guiding efficient sizing and cost-effective design of complex crossflow exchangers.

James et al. [60] conducted an uncertainty analysis on crossflow heat exchanger performance predictions. Their study aimed to quantify how uncertainties in input parameters stemming from experimental measurements and predictive models affect calculated parameters, such as heat transfer rate, effectiveness, and pressure drop. Using statistical uncertainty propagation, they showed that performance predictions can deviate significantly when uncertainties are correctly accounted for. This work emphasizes the need to incorporate uncertainty analysis to improve the reliability of heat exchanger design and performance evaluation.

Ataer et al. [61] developed three prediction approaches to model the transient thermal behavior of finned-tube, liquid/gas crossflow heat exchangers subject to step changes in inlet conditions. The methodologies included a lumped-capacitance model, a finite-difference numerical simulation, and a semi-analytical approach. The key parameters studied were thermal capacitance, fin geometry, fluid mass flow rates, and heat transfer coefficients. Results showed that both the fin and tube thermal masses significantly influence response time; accurate dynamic models must include fin conduction and fluid-side transient effects.

Stoitchkov and Dimitrova [62] developed a simplified method for calculating the effectiveness of crossflow plate heat exchangers utilized in indirect evaporative cooling systems. By introducing a correction factor to account for wet-surface conditions, the study used an ε -NTU approach to produce simpler performance charts. The results demonstrated that this method accurately predicts the effectiveness of heat exchangers across various inlet conditions. This advancement enables the design of compact indirect evaporative cooling units, thereby enhancing their reliability and reducing computational complexity.

Harris et al. [63] designed and fabricated a crossflow micro heat exchanger specifically for high heat-flux liquid-to-gas applications. Their methodology employed microfabrication techniques to create microchannels with hydraulic diameters of approximately 200 μm . The thermal performance was characterized using water/glycol and air flows. The experimental results showed a volumetric heat transfer coefficient exceeding 10 $\text{kW/m}^3\cdot\text{K}$ while maintaining a low air-side pressure drop. This study demonstrated that micro-scale crossflow exchangers are a viable option for

compact, high-performance thermal management systems.

Guo et al. [64] proposed and experimentally validated the uniformity principle of the temperature difference field in heat exchangers. Their study developed a theoretical framework demonstrating that achieving a uniform distribution of temperature differences maximizes overall performance. Using experimental data from prototype exchangers, the authors confirmed that designs with minor local deviations in temperature difference yield higher effectiveness and lower entropy generation. These findings provide essential design guidelines for enhancing thermal performance through uniform crossflow temperature distributions.

Crane and Jackson [65] developed optimization models for crossflow heat exchangers to enhance thermoelectric waste-heat recovery systems. The study employed both analytical and numerical methods to improve the geometry, flow configuration, and thermal-hydraulic performance of the heat exchangers. It accounted for key parameters, including heat exchanger size, pressure drop, heat duty, and thermoelectric module integration. The results showed significant performance improvements when the heat exchanger design was tailored explicitly for thermoelectric applications, indicating that optimized crossflow configurations can boost heat recovery efficiency in industrial systems.

Navarro and Cabezas-Gómez [66] introduced a new numerical method for calculating the thermal performance of crossflow heat exchangers. This approach divides the heat exchanger into a series of “one-pass mixed-unmixed” elements and applies energy balance and NTU-effectiveness relationships to each element. Validation against analytical solutions for one- to four-row configurations demonstrated very low error rates. This method allows for the prediction of effectiveness in complex configurations that have not been previously addressed in the literature.

Carluccio et al. [67] used it for cooling high-pressure oil in vehicle hydraulic systems. They utilized CFD simulations under realistic boundary conditions to evaluate the heat transfer and pressure drop characteristics across various fin geometries on both the air-side and oil-side. The results indicated deviations of up to $\pm 20\%$ from standard correlations, underscoring the importance of precise fin design and accurate simulations for predicting the thermal-hydraulic performance of compact heat exchangers.

Noie [68] investigated the thermal performance of an air-to-air thermosyphon heat exchanger using an ε -NTU model. The device consisted of 90 plate-finned copper thermosyphons arranged in six rows. Experiments were conducted with air face velocities ranging from approximately 0.5 to 5.5 m/s and hot-air inlet temperatures between 100 and 250 °C. The overall effectiveness of the system ranged from approximately 37% to 65%, with the lowest effectiveness observed when the heat capacity rates of both air streams were equal.

Navarro and Cabezas-Gómez [69] developed a mathematical model to calculate the effectiveness-NTU relationships for crossflow heat exchangers with complex flow arrangements. The model uses a tube-element discretization and iterative temperature-solving methods, enabling it to predict the performance of both mixed and unmixed configurations accurately. Validation against analytical results for single-pass systems showed deviations of less than 10%. Additionally, the model generated ε -NTU curves for multi-row and multi-pass exchangers, making it a practical tool for designing and evaluating the performance of non-standard crossflow heat exchanger geometries.

Mishra et al. [70] employed a genetic algorithm (GA) to optimize the second law of thermodynamics for a crossflow plate-fin heat exchanger with offset-strip fins; the optimization aimed to minimize entropy generation while satisfying specific heat load and compactness constraints. The design variables included fin geometry and exchanger core dimensions. The optimized configuration demonstrated a reduction of up to 22% in entropy generation compared to baseline designs, with only marginal increases in core volume. This study confirmed the effectiveness of GA for the thermodynamic optimization of compact heat exchangers.

Cabezas-Gómez et al. [71] evaluated the thermal performance of a crossflow heat exchanger with an innovative flow configuration designed to enhance heat transfer effectiveness and compactness. The study utilized both experimental measurements and numerical simulations across a range of Re and fluid temperatures. The performance metrics, such as the heat transfer coefficient, pressure drop, and effectiveness (using the ε -NTU method), were thoroughly analyzed. The results indicated that the new flow arrangement improved thermal performance by up to 20% compared to conventional designs, with only a slight increase in pressure drop. This research offers valuable insights for optimizing compact heat exchangers in HVAC and industrial applications.

Rao and Patel [72] employed a particle swarm optimization (PSO) algorithm to optimize the thermodynamic performance of a crossflow plate-fin heat exchanger. The study aimed to minimize entropy generation, volume, and annual costs by adjusting fin geometry and flow parameters. The results indicated that the PSO algorithm achieved optimal configurations, leading to an 18% reduction in total entropy generation and a 12% reduction in volume compared to baseline designs. Furthermore, the outcomes were closely aligned with benchmarks from GA, demonstrating the efficiency and robustness of PSO for optimizing heat exchangers.

Yoladi et al. [73] conducted an experimental evaluation of low-heat-exchanger systems equipped with helical fins. They utilized Response Surface Methodology (RSM) and Artificial Neural Network (ANN) modeling to assess both thermal and hydraulic performance. The researchers varied the fin geometry and flow conditions across a broad range. The results demonstrated an increase in heat transfer compared to conventional fins, with improvements of

approximately 30–45%, while maintaining acceptable pressure drop penalties. Additionally, their RSM and ANN models predicted performance with high accuracy, exhibiting low error rates, making them useful design tools for helical-finned exchangers.

Optimization and data-driven approaches provide a systematic framework for balancing the trade-off between heat transfer enhancement and hydraulic performance. Methods such as EGM, GA, PSO, and machine learning models help identify optimal design configurations that are difficult to obtain through conventional parametric studies. These approaches consistently improve thermo-hydraulic performance, often reflected in higher PEC values and lower entropy generation. A key advantage of these methods is their ability to integrate multiple design variables and constraints simultaneously. However, their effectiveness depends on the availability of high-quality experimental or numerical data for validation, and their implementation may require significant computational resources. These techniques are most appropriate for advanced design and optimization stages, particularly in complex systems where multiple competing parameters must be balanced.

Table 2 summarizes significant studies on crossflow and compact heat exchangers. It includes details such as the publication year, objectives, methodologies, key parameters, and main findings of each study. This structured overview facilitates a clear comparison of experimental, numerical, and theoretical approaches. Additionally, it highlights the evolution of enhancement techniques and the performance trends observed across various configurations. Table 2 also helps identify knowledge gaps that encourage further research.

Table 2. Summary of heat exchanger enhancement techniques: A comparative overview

Ref.	Objective	Methodology	Config.	Fluid Type	Working Conditions (Re, Pr)	Thermo-Hydraulic Evaluation (PEC/ j - f)
[6]	Tube-bank correlations	Exp	Inline/Staggered tubes	Air	$Re \approx 10^3$ – 10^5 , $Pr \approx 0.7$	Baseline correlations (reference)
[7]	Effect of spacing	Exp	Parallel plates	Air	$Re \approx 10^2$ – 10^4 , $Pr \approx 0.7$	Heat $\uparrow$, $\Delta P \uparrow \rightarrow$ $PEC < 1$
[10]	Boiling geometry	Exp	Tube bundle	Water/steam	$Re \approx 10^3$ – 10^4 , Pr variable	Geometry sensitive performance
[11]	Tube inclination	Exp	Inclined tubes	Air	$Re \approx 10^4$ – 10^5 , $Pr \approx 0.7$	Moderate PEC (≈ 1)
[13]	Offset strip fins	Exp	Offset fins	Air	$Re \approx 10^2$ – 10^4 , $Pr \approx 0.7$	Balanced $j/f \rightarrow$ $PEC \approx 1$ – 1.2
[14]	Wavy fins	Exp	Wavy fin-tube	Air	$Re \approx 500$ – 5000 , $Pr \approx 0.7$	$Nu \uparrow$, $\Delta P \uparrow \rightarrow$ $PEC \approx 1$
[21]	Wavy fin HX	Exp	Wavy fins	Air	$Re \leq 3000$, $Pr \approx 0.7$	$PEC \approx 1$
[22]	Louvered fins	Exp	Multi-louvered	Air	$Re \approx 400$ – 3000 , $Pr \approx 0.7$	$PEC > 1$
[23]	Elliptic tubes	Numerical	Elliptic bundle	Air	$Re \approx 5500$ – 14500 , $Pr \approx 0.7$	High PEC (> 1.5)
[24]	Surface profile	Exp	Corrugated plates	Air	$Re \approx 500$ – 5000 , $Pr \approx 0.7$	Moderate PEC (≈ 1)
[26]	PCHE channels	Numerical	Zigzag channels	Air/ CO_2	$Re \approx 10^3$ – 10^5 , $Pr \approx 0.7$ – 1	Optimal j/f for rectangular
[28]	Multi-pass HX	Numerical	Multi-passage	Air-water	$Re \approx 1000$ – 5000 , $Pr \approx 0.7$ – 7	$PEC > 1$

Continued on next page

Ref.	Objective	Methodology	Config.	Fluid Type	Working Conditions (Re, Pr)	Thermo-Hydraulic Evaluation (PEC/ j - f)
[29]	Protruded fins	Exp + CFD	Ellipsoidal fins	Air	Re $\approx$ 1000–5000, Pr $\approx$ 0.7	PEC > 1
[31]	Flattened tubes	Exp	Flat tube HX	Air	Re $\approx$ 2000–10000, Pr $\approx$ 0.7	PEC > 1
[32]	Wavy + elliptic	Exp + CFD	Waffle fins	Air	Re $\approx$ 1000–8000, Pr $\approx$ 0.7	Moderate PEC
[33]	Dimple fins	Numerical	Dimple fins	Air	Re = 1500–5000, Pr $\approx$ 0.7	PEC $\approx$ 1.16
[34]	Helix-wire fins	Exp	Helical fins	Air	Re $\approx$ 3000–20000, Pr $\approx$ 0.7	PEC > 1
[35]	Variable fins	Exp + CFD	Variable fins	Air	Re = 5000–25000, Pr $\approx$ 0.7	PEC > 1
[36]	Jet crossflow	Exp	Jet arrays	Air	Re = 1000–21000, Pr $\approx$ 0.7	Case dependent PEC
[38]	Baffles	Exp	Perforated baffles	Air	Re $\approx$ 5000–30000, Pr $\approx$ 0.7	High $\Delta P \rightarrow$ PEC $\approx$ 1
[39]	Vortex generators	Numerical	Winglets	Air	Re $\approx$ 3000–15000, Pr $\approx$ 0.7	PEC > 1
[40]	VG compact HX	Exp	Winglet VG	Air	Re $\approx$ 500–3000, Pr $\approx$ 0.7	PEC > 1
[41]	Multi-row VG	Exp	VG arrays	Air	Re = 220–960, Pr $\approx$ 0.7	PEC $\approx$ 1
[43]	Turbulators	Exp	Twisted rings	Air	Re $\approx$ 6000–20000, Pr $\approx$ 0.7	PEC $\approx$ 1.24
[45]	Vibration HX	Exp	Fin-tube	Air	Re $\approx$ 2000–10000, Pr $\approx$ 0.7	PEC > 1
[46]	Vibrating channel	Exp + CFD	Rectangular	Air	Re $\approx$ 2000–10000, Pr $\approx$ 0.7	PEC > 1
[47]	Wavy VG	Exp + CFD	Winglets	Air	Re = 500–2500, Pr $\approx$ 0.7	Moderate PEC
[48]	FIV effect	Numerical	Tube bundle	Air	5×10^3 – 3×10^4 , Pr $\approx$ 0.7	Limited PEC
[49]	Marine vibration	Exp	Plate HX	Water/air	Re $\approx$ 1000–10000	PEC > 1

Continued on next page

Ref.	Objective	Methodology	Config.	Fluid Type	Working Conditions (Re, Pr)	Thermo-Hydraulic Evaluation (PEC/ j - f)
[50]	Splitter plates	Numerical	Inclined plates	Air	Re = 5500–14500, Pr $\approx$ 0.7	High PEC
[51]	Porous cooling	Theoretical	Honeycomb	Air-water	Re $\approx$ 100–2000, Pr $\approx$ 0.7–7	PEC < 1 (ΔP dominated)
[53]	Ceramic HX	CFD	Plate-fin	Air	Re = 500–1500, Pr $\approx$ 0.7	Stable j - f
[54]	Nanofluids	Exp	Radiator	Water + CuO	Re = 50–1000, Pr $\approx$ 4–7	PEC > 1
[55]	Nanofluids	Exp	Radiator	Water + Al ₂ O ₃	Re $\approx$ 500–3000	PEC > 1
[56]	Graphene nanofluid	Exp	Radiator	Water-EG + graphene	Re $\approx$ 1000–10000	High PEC
[57]	Entropy method	Theoretical	Compact HX	General	General	Optimal PEC via EGM
[58]	FEM modeling	Numerical	Crossflow HX	Air/water	Re $\approx$ 1000–10000	Optimized PEC
[59]	ε -NTU method	Analytical	Multi-pass HX	General	General	Design optimization
[63]	Micro HX	Exp	Microchannel	Water/air	Re $\approx$ 100–2000	Very high performance
[64]	Uniformity principle	Theory + Exp	Crossflow HX	General	General	PEC maximized
[65]	Thermoelectric HX	Optimization	Crossflow HX	Gas	Re $\approx$ 1000–10000	Optimized PEC
[70]	GA optimization	Optimization	Plate-fin HX	Air	Re $\approx$ 2000–15000	PEC $\uparrow$
[72]	PSO optimization	Optimization	Plate-fin HX	Air	Re $\approx$ 2000–15000	PEC $\uparrow$
[73]	AI (ANN/RSM)	Exp + ML	Helical fins	Air	Re $\approx$ 3000–20000	High predictive PEC

Note: Config. = Configuration; HX = Heat exchanger; Re = Reynolds number; Pr = Prandtl number; PEC = Performance evaluation criterion; Nu = Nusselt number; ΔP = Pressure drop; CFD = Computational fluid dynamics; Exp = Experimental; VG = Vortex generators; GA = Genetic algorithm; PSO = Particle swarm optimization; AI = Artificial intelligence; ANN = Artificial Neural Network; RSM = Response Surface Methodology; ML = Machine learning.

4 Research Gaps Not Adequately Addressed

Despite the substantial progress reported in the literature, several critical gaps remain in the development and optimization of cross-flow heat exchangers. These gaps are identified from the limitations observed in the reviewed studies and are explicitly linked to the corresponding thematic sections and the comparative results summarized in Table 2.

A. Lack of a Unified Thermo-Hydraulic Evaluation Framework

Many studies on geometric modifications—such as wavy fins, corrugated surfaces, and louvered fins (e.g., Refs. [14, 21, 24])—report significant increases in heat transfer; however, these improvements are often presented without a consistent assessment of the associated pressure-drop penalty. As shown in Table 2, several configurations yield higher Nusselt numbers but only moderate or even suboptimal overall performance (PEC $\approx$ 1 or < 1). It indicates a lack of standardized evaluation criteria across studies. Therefore, a major gap remains in adopting unified performance metrics (e.g., PEC or j/f ratio) to enable meaningful comparison and optimization.

B. Limited Multi-Objective Optimization Across Design Parameters

Although optimization methods such as EGM and evolutionary algorithms (Refs. [57, 70, 72]) have been applied, many studies still rely on single-parameter or single-objective optimization, particularly in geometric modification and flow control sections. For instance, studies of VG and baffles (Refs. [38, 39]) often prioritize heat transfer enhancement without fully optimizing the trade-off with pressure drop. This highlights a gap in the application of comprehensive multi-objective optimization frameworks that simultaneously consider thermal performance, hydraulic losses, and energy efficiency.

C. Insufficient Integration of Hybrid Enhancement Techniques

The reviewed literature indicates that most studies examine enhancement techniques in isolation. For example, geometric modifications (Section a) and advanced materials such as nanofluids or porous media (Section c; Refs. [51, 54–56]) are rarely combined systematically. However, Table 2 suggests that hybrid approaches could yield higher thermo-hydraulic performance ($PEC > 1$) by leveraging multiple enhancement mechanisms simultaneously. The lack of studies integrating geometry, flow control, and advanced materials within a unified framework represents a significant research gap.

D. Narrow Operating Ranges and Lack of Generalized Correlations

A large portion of experimental studies (e.g., Refs. [21, 22, 33]) are conducted within relatively narrow Re ranges (typically $Re < 5,000$) and with specific working fluids (mostly air or water). This limitation restricts the generalizability of the findings. As shown in Table 2, Re and Pr variations are not consistently explored, limiting the applicability of the proposed correlations across different operating conditions. Therefore, there is a need for studies that cover wider Re–Pr ranges and develop generalized predictive correlations.

E. Pressure Drop Penalty in Advanced Materials and Porous Media

Although advanced materials such as porous media and nanofluids demonstrate substantial heat-transfer enhancement (Refs. [52, 55–57]), they often introduce significant hydraulic resistance. As shown in Table 2, porous media-based systems may reduce overall efficiency ($PEC < 1$) due to excessive pressure drop. This highlights a key limitation in balancing thermal gains with flow resistance and underscores the need for optimized porous structures and flow management strategies.

F. Limited Application of Data-Driven and Machine Learning Approaches

Recent studies have begun to explore machine learning and data-driven techniques (e.g., Ref. [73]); however, their application remains limited compared with traditional methods. While these approaches show strong potential for performance prediction and optimization, they are not yet widely integrated into experimental and CFD-based studies. This highlights a gap in developing hybrid ML–physics-based models capable of accurately predicting thermo-hydraulic performance across a wide range of configurations.

G. Lack of Long-Term Performance and Practical Validation

Most of the reviewed studies focus on short-term experimental or numerical analyses conducted under controlled conditions. There is limited investigation into long-term performance, fouling, material degradation, and real-world operating environments. This is particularly evident in studies involving nanofluids and advanced materials, where stability and durability remain critical concerns. Therefore, further research is needed to validate these enhancement techniques under realistic industrial conditions.

5 Conclusion

This study presents a structured, comparative review of cross-flow heat exchangers, covering both classical developments and modern enhancement techniques, including geometric modifications, flow control, advanced materials, and optimization/data-driven approaches. The analysis, supported by a summary table above, shows that overall thermo-hydraulic performance is governed by a trade-off between heat transfer enhancement and pressure drop, which should be evaluated using unified metrics such as the PEC or the j/f ratio.

From an engineering standpoint, several practical insights can be drawn. First, enhancement techniques should not be selected based solely on heat transfer improvement; instead, a balanced assessment of thermal and hydraulic performance is essential. Geometric modifications, such as louvered and elliptical fins, generally provide reliable performance at moderate to high Re, while flow control techniques (e.g., VG) are particularly effective in low-to-intermediate Re because they promote mixing with controlled pressure loss. In contrast, advanced materials, including porous media and nanofluids, offer substantial thermal enhancement but may introduce significant pressure-drop penalties or operational challenges, limiting their applicability in systems where pumping power is critical.

Furthermore, optimization and data-driven methods, including multi-objective algorithms and machine learning models, have emerged as powerful tools for identifying optimal configurations and managing competing design parameters. These approaches are particularly valuable in complex systems where traditional design methods are insufficient. The growing trend toward hybrid enhancement strategies that combine multiple techniques indicates a promising direction for achieving superior thermo-hydraulic performance.

In practical applications, the selection and design of cross-flow heat exchangers should consider operating conditions, system constraints, and performance objectives. Compact systems may benefit from high-performance

enhancement techniques despite increased pressure drop, whereas large-scale systems often require energy-efficient designs with minimized hydraulic losses.

Future research should focus on developing standardized performance evaluation frameworks, expanding the applicability of hybrid and data-driven approaches, and validating enhancement techniques under realistic operating conditions. Such efforts will contribute to the development of more efficient, reliable, and application-specific cross-flow heat exchanger designs.

Author Contributions

Z.A.A.H. contributed to surveying the studies conducted on the research topic, while E.F.A. and A.A. reviewed and organized the article.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

References

- [1] R. K. Shah and D. P. Sekulić, *Fundamentals of Heat Exchanger Design*. Hoboken, NJ, USA: John Wiley & Sons, Inc., 2003. <https://doi.org/10.1002/9780470172605>
- [2] T. L. Bergman, A. S. Lavine, F. P. Incropera, and D. P. DeWitt, *Fundamentals of Heat and Mass Transfer, 7th ed.* Hoboken, NJ, USA: John Wiley & Sons, Inc., 2011.
- [3] W. M. Kays and A. L. London, *Compact Heat Exchangers, 3rd ed.* [S.l.]: Scientific International, 2018.
- [4] A. S. Mujumdar, "Heat exchanger design handbook T. Kuppan Marcel Dekker Inc., New York 2000, 1118 pages," *Dry. Technol.*, vol. 18, no. 9, pp. 2167–2168, 2000. <https://doi.org/10.1080/07373930008917833>
- [5] K. Thulukkanam, *Heat Exchanger Design Handbook, 2nd Edition*. Boca Raton: CRC Press, 2013. <https://doi.org/10.1201/b14877>
- [6] A. Žukauskas and R. Ulinskas, "Efficiency parameters for heat transfer in tube banks," *Heat Transf. Eng.*, vol. 6, no. 1, pp. 19–25, 1985. <https://doi.org/10.1080/01457638508939614>
- [7] M. Molki and C. M. Yuen, "Effect of interwall spacing on heat transfer and pressure drop in a corrugated-wall duct," *Int. J. Heat Mass Transf.*, vol. 29, no. 7, pp. 987–997, 1986. [https://doi.org/10.1016/0017-9310\(86\)90198-5](https://doi.org/10.1016/0017-9310(86)90198-5)
- [8] A. Žukauskas, "Heat transfer from tubes in crossflow," *Adv. Heat Transf.*, vol. 18, pp. 87–159, 1987. [https://doi.org/10.1016/S0065-2717\(08\)70118-7](https://doi.org/10.1016/S0065-2717(08)70118-7)
- [9] M. Monheit and J. Freim, "Effect of tube bank inclination on the thermal hydraulic performance of air-cooled heat exchangers," *Heat Transf. Eng.*, vol. 8, no. 1, pp. 19–24.
- [10] M. K. Jensen, M. J. Reinke, and J. T. Hsu, "The influence of tube bundle geometry on cross-flow boiling heat transfer and pressure drop," *Exp. Therm. Fluid Sci.*, vol. 2, no. 4, pp. 465–476, 1989. [https://doi.org/10.1016/0894-1777\(89\)90008-3](https://doi.org/10.1016/0894-1777(89)90008-3)
- [11] E. Achenbach, "Heat transfer from a staggered tube bundle in cross-flow at high Reynolds numbers," *Int. J. Heat Mass Transf.*, vol. 32, no. 2, pp. 271–280, 1989. [https://doi.org/10.1016/0017-9310\(89\)90174-9](https://doi.org/10.1016/0017-9310(89)90174-9)
- [12] A. Nir, "Heat transfer and friction factor correlations for crossflow over staggered finned tube banks," *Heat Transf. Eng.*, vol. 12, no. 1, pp. 43–58.
- [13] R. M. Manglik and A. E. Bergles, "Heat transfer and pressure drop correlations for the rectangular offset strip fin compact heat exchanger," *Exp. Therm. Fluid Sci.*, vol. 10, no. 2, pp. 171–180, 1995. [https://doi.org/10.1016/0894-1777\(94\)00096-Q](https://doi.org/10.1016/0894-1777(94)00096-Q)
- [14] Z. Wang, R. Wang, Z. Li, M. Wang, and L. Wan, "Numerical investigation on the effect of cylindrical turbulator on performance of corrugated plate-fin heat exchanger," *Appl. Therm. Eng.*, vol. 230, p. 120726, 2023. <https://doi.org/10.1016/j.applthermaleng.2023.120726>
- [15] A. R. Martin, C. Saltiel, and W. Shyy, "Frictional losses and convective heat transfer in sparse, periodic cylinder arrays in cross flow," *Int. J. Heat Mass Transf.*, vol. 41, no. 15, pp. 2383–2397, 1998. [https://doi.org/10.1016/S0017-9310\(97\)00300-1](https://doi.org/10.1016/S0017-9310(97)00300-1)
- [16] C. F. Lange, F. Durst, and M. Breuer, "Momentum and heat transfer from cylinders in laminar crossflow at $10^{-4} \leq Re \leq 200$," *Int. J. Heat Mass Transf.*, vol. 41, no. 22, pp. 3409–3430, 1998. [https://doi.org/10.1016/S0017-9310\(98\)00077-5](https://doi.org/10.1016/S0017-9310(98)00077-5)

- [17] H. Li and V. Kottke, "Visualization and determination of local heat transfer coefficients in shell-and-tube heat exchangers for staggered tube arrangement by mass transfer measurements," *Exp. Therm. Fluid Sci.*, vol. 17, no. 3, pp. 210–216, 1998. [https://doi.org/10.1016/S0894-1777\(97\)10064-4](https://doi.org/10.1016/S0894-1777(97)10064-4)
- [18] C. Ranganayakulu and K. N. Seetharamu, "The combined effects of longitudinal heat conduction, flow nonuniformity and temperature nonuniformity in crossflow plate-fin heat exchangers," *Int. Commun. Heat Mass Transf.*, vol. 26, no. 5, pp. 669–678, 1999. [https://doi.org/10.1016/S0735-1933\(99\)00053-6](https://doi.org/10.1016/S0735-1933(99)00053-6)
- [19] B. Yu and W. Q. Tao, "Pressure drop and heat transfer characteristics of turbulent flow in annular tubes with internal wave-like longitudinal fins," *Heat Mass Transf.*, vol. 40, pp. 643–651, 2004. <https://doi.org/10.1007/s00231-003-0449-5>
- [20] A. Erek, B. Özerdem, L. Bilir, and Z. Ilken, "Effect of geometrical parameters on heat transfer and pressure drop characteristics of plate fin and tube heat exchangers," *Appl. Therm. Eng.*, vol. 25, no. 14–15, pp. 2421–2431, 2005. <https://doi.org/10.1016/j.applthermaleng.2004.12.019>
- [21] D. Junqi, C. Jiangping, C. Zhijiu, Z. Yimin, and Z. Wenfeng, "Heat transfer and pressure drop correlations for the wavy fin and flat tube heat exchangers," *Appl. Therm. Eng.*, vol. 27, no. 11–12, pp. 2066–2073, 2007. <https://doi.org/10.1016/j.applthermaleng.2006.11.012>
- [22] J. Dong, J. Chen, Z. Chen, W. Zhang, and Y. Zhou, "Heat transfer and pressure drop correlations for the multi-louvered fin compact heat exchangers," *Energy Convers. Manag.*, vol. 48, no. 5, pp. 1506–1515, 2007. <https://doi.org/10.1016/j.enconman.2006.11.023>
- [23] T. A. Ibrahim and A. Gomaa, "Thermal performance criteria of elliptic tube bundle in crossflow," *Int. J. Therm. Sci.*, vol. 48, no. 11, pp. 2148–2158, 2009. <https://doi.org/10.1016/j.ijthermalsci.2009.03.011>
- [24] A. Durmuş, H. Benli, I. Kurtbaş, and H. Gül, "Investigation of heat transfer and pressure drop in plate heat exchangers having different surface profiles," *Int. J. Heat Mass Transf.*, vol. 52, no. 5–6, pp. 1451–1457, 2009. <https://doi.org/10.1016/j.ijheatmasstransfer.2008.07.052>
- [25] D. Dović, B. Palm, and S. Švaić, "Generalized correlations for predicting heat transfer and pressure drop in plate heat exchanger channels of arbitrary geometry," *Int. J. Heat Mass Transf.*, vol. 52, no. 19–20, pp. 4553–4563, 2009. <https://doi.org/10.1016/j.ijheatmasstransfer.2009.03.074>
- [26] S. M. Lee and K. Y. Kim, "Comparative study on performance of a zigzag printed circuit heat exchanger with various channel shapes and configurations," *Heat Mass Transf.*, pp. 1021–1028, 2013. <https://doi.org/10.1007/s00231-013-1149-4>
- [27] Y. Y. Liang, C. C. Liu, C. Z. Li, and J. P. Chen, "Experimental and simulation study on the air side thermal hydraulic performance of automotive heat exchangers," *Appl. Therm. Eng.*, vol. 87, pp. 305–315, 2015. <https://doi.org/10.1016/j.applthermaleng.2015.05.018>
- [28] M. Khaled, K. Faraj, H. El Hage, J. Faraj, R. Taher, and M. Mortazavi, "Multi-passage concept applied to water-air cross flow tubes-and-fins heat exchangers—Thermal modelling and feasibility study," *Int. J. Thermofluids*, vol. 17, p. 100291, 2023. <https://doi.org/10.1016/j.ijft.2023.100291>
- [29] L. Zhang, J. Dong, S. Sun, and Z. Chen, "Numerical simulation and sensitivity analysis on an improved Trombe wall," *Sustain. Energy Technol. Assess.*, vol. 43, p. 100941, 2021. <https://doi.org/10.1016/j.seta.2020.100941>
- [30] W. Wu, J. Luo, D. Li, X. Feng, L. Tang, Z. Fang, and Z. Zheng, "Experimental investigation of heat transfer performance of a finned-tube heat exchanger under frosting conditions," *Sustain. Cities Soc.*, vol. 80, p. 103752, 2022. <https://doi.org/10.1016/j.scs.2022.103752>
- [31] A. Rukruang, T. Chittiphalsri, N. Chimres, J. Kaew-On, and S. Wongwises, "Experimental investigation on the condensation of R32 flowing inside alternating cross-section flattened tubes with different aspect ratios," *Int. J. Heat Mass Transf.*, vol. 214, p. 124397, 2023. <https://doi.org/10.1016/j.ijheatmasstransfer.2023.124397>
- [32] Y. T. Lee, L. H. Chien, J. He, C. Y. Wen, and A. S. Yang, "Air side performance characterization of wavy fin-and-tube heat exchangers having elliptic tubes with large waffle heights," *Appl. Therm. Eng.*, vol. 217, p. 119220, 2022. <https://doi.org/10.1016/j.applthermaleng.2022.119220>
- [33] G. Song, E. S. Moon, J. J. Park, S. M. Song, and S. J. Yook, "Performance investigation of radial heat sink with pin fins placed normally on circular base edge," *Int. J. Therm. Sci.*, vol. 171, p. 107187, 2022. <https://doi.org/10.1016/j.ijthermalsci.2021.107187>
- [34] Y. Maghsoudali, A. Rastegarkoutenaeei, M. Sahami, and M. G. Bandy, "Experimental investigating the thermal and hydraulic performance of heat exchangers with helix wire finned tube," *Appl. Therm. Eng.*, vol. 231, p. 120919, 2023. <https://doi.org/10.1016/j.applthermaleng.2023.120919>
- [35] L. Liu, X. Jiang, H. Tang, H. Xu, and X. Zhang, "Experimental and numerical investigation on the flow and heat transfer characteristics of the variable cross-section internally finned tube," *Int. J. Heat Mass Transf.*, vol. 239, p. 126589, 2025. <https://doi.org/10.1016/j.ijheatmasstransfer.2024.126589>
- [36] N. T. Obot and T. A. Trabold, "Impingement heat transfer within arrays of circular jets: Part 1—Effects of minimum, intermediate, and complete crossflow for small and large spacings," *J. Heat Transf.*, vol. 109, no. 4,

pp. 872–879, 1987. <https://doi.org/10.1115/1.3248197>

- [37] C. Ranganayakulu, K. N. Seetharamu, and K. V. Sreevatsan, “The effects of inlet fluid flow nonuniformity on thermal performance and pressure drops in crossflow plate-fin compact heat exchangers,” *Int. J. Heat Mass Transf.*, vol. 40, no. 1, pp. 27–38, 1996. [https://doi.org/10.1016/S0017-9310\(96\)00087-7](https://doi.org/10.1016/S0017-9310(96)00087-7)
- [38] P. Dutta and S. Dutta, “Effect of baffle size, perforation, and orientation on internal heat transfer enhancement,” *Int. J. Heat Mass Transf.*, vol. 41, no. 19, pp. 3005–3013, 1998. [https://doi.org/10.1016/S0017-9310\(98\)00016-7](https://doi.org/10.1016/S0017-9310(98)00016-7)
- [39] S. Tiwari, D. Maurya, G. Biswas, and V. Eswaran, “Heat transfer enhancement in cross-flow heat exchangers using oval tubes and multiple delta winglets,” *Int. J. Heat Mass Transf.*, vol. 46, no. 15, pp. 2841–2856, 2003. [https://doi.org/10.1016/S0017-9310\(03\)00047-4](https://doi.org/10.1016/S0017-9310(03)00047-4)
- [40] A. Joardar and A. M. Jacobi, “Impact of leading edge delta-wing vortex generators on the thermal performance of a flat tube, louvered-fin compact heat exchanger,” *Int. J. Heat Mass Transf.*, vol. 48, no. 8, pp. 1480–1493, 2005. <https://doi.org/10.1016/j.ijheatmasstransfer.2004.10.018>
- [41] A. Joardar and A. M. Jacobi, “Heat transfer enhancement by winglet-type vortex generator arrays in compact plain-fin-and-tube heat exchangers,” *Int. J. Refrig.*, vol. 31, no. 1, pp. 87–97, 2008. <https://doi.org/10.1016/j.ijrefrig.2007.04.011>
- [42] S. Wang, J. Wen, and Y. Li, “An experimental investigation of heat transfer enhancement for a shell-and-tube heat exchanger,” *Appl. Therm. Eng.*, vol. 29, no. 11–12, pp. 2433–2438, 2009. <https://doi.org/10.1016/j.applthermaleng.2008.12.008>
- [43] C. Thianpong, K. Yongsiri, K. Nanan, and S. Eiamsa-ard, “Thermal performance evaluation of heat exchangers fitted with twisted-ring turbulators,” *Int. Commun. Heat Mass Transf.*, vol. 39, no. 6, pp. 861–868, 2012. <https://doi.org/10.1016/j.icheatmasstransfer.2012.04.004>
- [44] M. Hanuszkiewicz-Drapala, T. Bury, and K. Widziewicz, “Numerical model of a cross-flow heat exchanger with non-uniform flow of media,” *J. Power Technol.*, vol. 93, no. 5, pp. 295–302, 2013.
- [45] D. Li, X. Yang, S. Wang, D. Duan, Z. Wan, G. Xia, and W. Liu, “Experimental research on vibration-enhanced heat transfer of fin-tube vehicle radiator,” *Appl. Therm. Eng.*, vol. 180, p. 115836, 2020. <https://doi.org/10.1016/j.applthermaleng.2020.115836>
- [46] J. Fu, X. Miao, Q. Zuo, H. Tang, Y. Li, Y. Zhang, and B. Sunden, “Heat transfer and field synergy characteristics in a rectangular unit channel under mechanical vibration,” *Int. Commun. Heat Mass Transf.*, vol. 136, p. 106176, 2022. <https://doi.org/10.1016/j.icheatmasstransfer.2022.106176>
- [47] K. W. Song, K. Sun, X. J. Lu, Q. F. Gao, Q. Z. Hou, and B. D. Gu, “Effect of wavy delta winglet vortex generators on heat transfer performance of a fin-and-tube heat exchanger,” *Int. J. Heat Fluid Flow*, vol. 108, p. 109485, 2024. <https://doi.org/10.1016/j.ijheatfluidflow.2024.109485>
- [48] H. Zhao, P. Gao, X. Li, R. Tian, H. Wei, and S. Tan, “The interaction between cross-flow induced vibration and convection heat transfer in tube bundle at subcritical Reynolds number,” *Appl. Therm. Eng.*, vol. 236, p. 121920, 2024. <https://doi.org/10.1016/j.applthermaleng.2023.121920>
- [49] Y. Wang, S. Ding, A. Yan, H. Miao, F. Wang, and J. Yuan, “Flow and heat transfer performance analysis of brazed plate heat exchangers under marine vibration conditions,” *Int. J. Therm. Sci.*, vol. 205, p. 109270, 2024. <https://doi.org/10.1016/j.ijthermalsci.2024.109270>
- [50] S. Kaushik, H. Singh, P. R. Pati, J. Giri, T. Sathish, M. Kanan, S. Chamoli, and C. Kishore, “Thermal performance and flow control in cross-flow heat exchangers using inclined splitter plate,” *Int. J. Thermofluids*, vol. 29, p. 101351, 2025. <https://doi.org/10.1016/j.ijft.2025.101351>
- [51] Y. J. Dai and K. Sumathy, “Theoretical study on a cross-flow direct evaporative cooler using honeycomb paper as packing material,” *Appl. Therm. Eng.*, vol. 22, no. 13, pp. 1417–1430, 2002. [https://doi.org/10.1016/S1359-4311\(02\)00069-8](https://doi.org/10.1016/S1359-4311(02)00069-8)
- [52] C. Zhan, X. Zhao, S. Smith, and S. B. Riffat, “Numerical study of a M-cycle cross-flow heat exchanger for indirect evaporative cooling,” *Build. Environ.*, vol. 46, no. 3, pp. 657–668, 2011. <https://doi.org/10.1016/j.buildenv.2010.09.011>
- [53] D. B. Monteiro and P. E. B. de Mello, “Thermal performance and pressure drop in a ceramic heat exchanger evaluated using CFD simulations,” *Energy*, vol. 45, no. 1, pp. 489–496, 2012. <https://doi.org/10.1016/j.energy.2012.02.012>
- [54] S. M. Peyghambarzadeh, S. H. Hashemabadi, M. Naraki, and Y. Vermahmoudi, “Experimental study of overall heat transfer coefficient in the application of dilute nanofluids in the car radiator,” *Appl. Therm. Eng.*, vol. 52, no. 1, pp. 8–16, 2013. <https://doi.org/10.1016/j.applthermaleng.2012.11.013>
- [55] D. R. Ray and D. K. Das, “Superior performance of nanofluids in an automotive radiator,” *J. Therm. Sci. Eng. Appl.*, vol. 6, no. 4, 2014. <https://doi.org/10.1115/1.4027302>
- [56] C. Selvam, D. Mohan Lal, and S. Harish, “Enhanced heat transfer performance of an automobile radiator with

- graphene based suspensions,” *Appl. Therm. Eng.*, vol. 123, pp. 50–60, 2017. <https://doi.org/10.1016/j.applthermaleng.2017.05.076>
- [57] D. P. Sekulić and C. V. Herman, “One approach to irreversibility minimization in compact crossflow heat exchanger design,” *Int. Commun. Heat Mass Transf.*, vol. 13, no. 1, pp. 23–32, 1986. [https://doi.org/10.1016/0735-1933\(86\)90069-2](https://doi.org/10.1016/0735-1933(86)90069-2)
- [58] S. G. Ravikumaur, K. N. Seetharamu, and P. A. A. Narayana, “Performance evaluation of crossflow compact heat exchangers using finite elements,” *Int. J. Heat Mass Transf.*, vol. 32, no. 5, pp. 889–894, 1989. [https://doi.org/10.1016/0017-9310\(89\)90238-X](https://doi.org/10.1016/0017-9310(89)90238-X)
- [59] R. K. Shah and A. Pignotti, “Thermal analysis of complex crossflow exchangers in terms of standard configurations,” *J. Heat Transf.*, vol. 115, no. 2, 1993. <https://doi.org/10.1115/1.2910686>
- [60] C. A. James, R. P. Taylor, and B. K. Hodge, “The application of uncertainty analysis to cross-flow heat exchanger performance predictions,” *Heat Transf. Eng.*, vol. 16, no. 4, pp. 50–62, 1995. <https://doi.org/10.1080/01457639508939863>
- [61] Ö. E. Ataer, A. Ileri, and Y. Göğüş, “Transient behaviour of finned-tube cross-flow heat exchangers,” *Int. J. Refrig.*, vol. 18, no. 3, pp. 153–160, 1995. [https://doi.org/10.1016/0140-7007\(94\)00002-F](https://doi.org/10.1016/0140-7007(94)00002-F)
- [62] N. J. Stoitchkov and G. I. Dimitrov, “Effectiveness of crossflow plate heat exchanger for indirect evaporative cooling,” *Int. J. Refrig.*, vol. 21, no. 6, pp. 463–471, 1998. [https://doi.org/10.1016/S0140-7007\(98\)00004-8](https://doi.org/10.1016/S0140-7007(98)00004-8)
- [63] C. Harris, M. Despa, and K. Kelly, “Design and fabrication of a cross flow micro heat exchanger,” *J. Microelectromech. Syst.*, vol. 9, no. 4, pp. 502–508, 2000. <https://doi.org/10.1109/84.896772>
- [64] Z. Y. Guo, S. Q. Zhou, Z. X. Li, and L. G. Chen, “Theoretical analysis and experimental confirmation of the uniformity principle of temperature difference field in heat exchanger,” *Int. J. Heat Mass Transf.*, vol. 45, no. 10, pp. 2119–2127, 2002. [https://doi.org/10.1016/S0017-9310\(01\)00297-6](https://doi.org/10.1016/S0017-9310(01)00297-6)
- [65] D. T. Crane and G. S. Jackson, “Optimization of cross flow heat exchangers for thermoelectric waste heat recovery,” *Energy Convers. Manag.*, vol. 45, no. 9–10, pp. 1565–1582, 2004. <https://doi.org/10.1016/j.enconman.2003.09.003>
- [66] H. A. Navarro and L. Cabezas-Gómez, “A new approach for thermal performance calculation of cross-flow heat exchangers,” *Int. J. Heat Mass Transf.*, vol. 48, no. 18, pp. 3880–3888, 2005. <https://doi.org/10.1016/j.ijheatmasstransfer.2005.03.027>
- [67] E. Carluccio, G. Starace, A. Ficarella, and D. Laforgia, “Numerical analysis of a cross-flow compact heat exchanger for vehicle applications,” *Appl. Therm. Eng.*, vol. 25, no. 13, pp. 1995–2013, 2005. <https://doi.org/10.1016/j.applthermaleng.2004.11.013>
- [68] S. H. Noie, “Investigation of thermal performance of an air-to-air thermosyphon heat exchanger using ϵ -NTU method,” *Appl. Therm. Eng.*, vol. 26, no. 5–6, pp. 559–567, 2006. <https://doi.org/10.1016/j.applthermaleng.2005.07.012>
- [69] H. A. Navarro and L. Cabezas-Gómez, “Effectiveness-NTU computation with a mathematical model for cross-flow heat exchangers,” *Braz. J. Chem. Eng.*, vol. 24, no. 4, pp. 509–521, 2007. <https://www.semanticscholar.org/reader/0b0893aa2fad9bbd160f2fa2e5d1aa5c92eb6b7b>
- [70] M. Mishra, P. K. Das, and S. Sarangi, “Effect of temperature and flow nonuniformity on transient behaviour of crossflow heat exchanger,” *Int. J. Heat Mass Transf.*, vol. 51, no. 9–10, pp. 2583–2592, 2008. <https://doi.org/10.1016/j.ijheatmasstransfer.2007.07.054>
- [71] L. Cabezas-Gómez, H. A. Navarro, S. M. de Godoy, A. Campo, and J. M. Saiz-Jabardo, “Thermal characterization of a cross-flow heat exchanger with a new flow arrangement,” *Int. J. Therm. Sci.*, vol. 48, no. 11, pp. 2165–2170, 2009. <https://doi.org/10.1016/j.ijthermalsci.2009.03.015>
- [72] R. V. Rao and V. K. Patel, “Thermodynamic optimization of cross flow plate-fin heat exchanger using a particle swarm optimization algorithm,” *Int. J. Therm. Sci.*, vol. 49, no. 9, pp. 1712–1721, 2010. <https://doi.org/10.1016/j.ijthermalsci.2010.04.001>
- [73] M. Yoladi, E. F. Akyurek, and İkotcioglu, “Experimental investigation of cross-flow heat exchangers with helical fins: Performance analysis via RSM and ANN,” *Int. J. Therm. Sci.*, vol. 218, p. 110111, 2025. <https://doi.org/10.1016/j.ijthermalsci.2025.110111>